

AI Agents as Tutors: Supporting Learning Without Replacing Teachers

Marcelo Vargas

James Madison University

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Dr. Jim Bywater

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Artificial Intelligence has become one of the biggest topics in education over the past few years. Many people either think AI will completely change education for the better or that it will replace teachers and encourage students to cheat. I think the reality is somewhere in the middle. Whether an AI agent is useful depends on how it is designed and how it is used. Based on my experience creating an Algebra 1 tutoring agent, I believe AI agents can be valuable learning tools when they are designed to support students rather than do the work for them. In a high school mathematics classroom, an AI tutor should act as an extension of the teacher by providing individualized guidance, helping students think through problems, and giving them opportunities to practice outside of class, while still keeping the teacher responsible for instruction and assessment.

For this project, I designed an AI agent to help Algebra 1 students prepare for a Unit 3 assessment covering equations and inequalities. Rather than creating an agent that simply solved problems or gave students the answers, I intentionally designed it to act like a tutor. The agent asks guiding questions, gives hints, provides feedback, and encourages students to explain their reasoning before moving forward. I spent time refining the prompts and testing different situations to make sure it behaved consistently. Whenever I found cases where the agent was giving away answers or not responding the way I intended, I adjusted the prompt until it better reflected the role of a teacher or tutor. That process showed me that creating an effective AI agent is much more than writing one prompt. It requires continuous testing, reflection, and revision to make sure the tool supports learning instead of replacing it.

One of the biggest strengths of this type of AI agent is its ability to provide individualized support whenever students need it. In a typical classroom, I may have thirty students, each struggling with different concepts. It is impossible for me to spend unlimited one-on-one time with every student. An AI tutor helps fill that gap by allowing students to practice at their own pace, ask questions without feeling embarrassed, and receive immediate feedback. Students who need additional support outside of school also benefit because they have access to tutoring

that they might not otherwise be able to afford. At the same time, the AI does not replace the teacher. The teacher still decides what students should learn, creates assessments, identifies misconceptions, and builds relationships with students (Kent & du Boulay, 2022, pp. 6-8). The AI simply extends the teacher's ability to support learning beyond the classroom.

Chapter 1 of AI for Learning reinforces this idea by arguing that good AI systems are built around the strengths of both humans and machines instead of trying to replace people (Kent & du Boulay, 2022, pp. 6-10). The authors explain that AI is better at tasks such as processing information quickly, recognizing patterns, and providing immediate responses, while humans are better at adapting to new situations, understanding context, exercising judgement, and building relationships (Kent & du Boulay, 2022, pp. 6-8). The goal is to create a partnership where each contributes what it does best instead of expecting AI to become a replacement for teachers.

This partnership closely matches how I envisioned my Algebra 1 tutoring agent. Chapter 1 discusses several different models of human-AI collaboration and emphasizes approaches that augment human abilities instead of replacing them. I believe my agent fits best within the dialogue and review levels of collaboration because the AI and student work together through conversation. The AI asks questions, provides feedback, and reviews the student's thinking, but does not complete the assignment for them. This keeps the student actively involved in the learning process while using AI as a source of guidance. The teacher also remains involved by deciding when and how students should use the agent, making the AI another instructional tool rather than the instructor itself (Kent & du Boulay, 2022, pp. 8-15).

The reading also discusses the learning affordances and constraints of AI systems (Kent & du Boulay, 2022, pp. 15-18). My experience building this agent reflected both. One affordance is personalization. The agent can adapt its responses based on what the student asks and provide explanations that are different from what might have been presented during class. It can also provide additional practice and immediate feedback at any time of day. However, I also

discover important limitations. Mathematics is often worked out on paper, and students do not always explain their thinking clearly through text. Because of that, the AI sometimes has difficulty identifying exactly where a student's misunderstanding begins. As the designer of the agent, I realized that these constraints are just as important as the affordances because they influence when AI should and should not be used (Kent & du Boulay, 2022, pp. 15-18). The teacher still needs to observe students solving problems, ask follow-up questions, and make instructional decisions that AI cannot reliably make.

Another lesson I learned from designing this project is that AI agents have value beyond student tutoring. Before this assignment, I mostly thought about AI as something teachers would personally use. Building an agent made me realize teachers can create specialized AI tools for other educators and students. A department could share tutoring agents for different units, lesson-planning assistants, grading support tools, or agents that help unpack curriculum standards. Instead of every teacher starting from scratch, schools could collaborate by building and improving agents together. That idea has significant potential because it allows teachers to share expertise while still maintaining professional judgement over how AI is used in their classrooms.

Overall, my experience designing an AI tutoring agent strengthened my belief that AI has an important place in education when it is used thoughtfully with 'humans in the loop'. AI should not replace teachers or remove students from the learning process. Instead, it should provide additional opportunities for practice, individualized support, and feedback while allowing teachers to focus on the parts of teaching that require human judgement, creativity and relationships. My Algebra 1 tutoring agent demonstrated that AI can be an effective educational partner when it encourages students to think rather than simply providing answers (Kent & du Boulay, 2022). As AI continues to improve, I believe the most successful classrooms will not be the ones that replace teachers with technology, but the ones that intentionally combine the strengths of both.

References

Kent, C., & du Boulay, B. (2022). *AI for Learning* (1st ed.). CRC Press.

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